

Faculty of Computer Applications and IT

Subject: Embedded Systems (230701302)

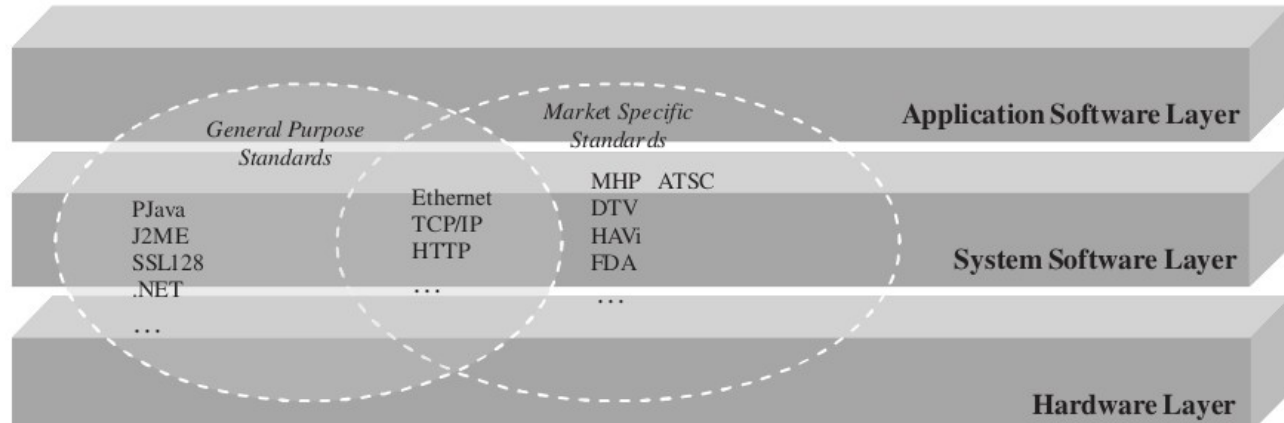
Chapter – 2 : Know Your Standards

Delivered by: Dr. Krupa Mehta

- This material is prepared for the students of MCA, FCAIT – PG Programme, GLS University considering the syllabus for the subject “Embedded Systems”
- The presentation included here outlines only the main points and key highlights of the course. It does not cover the entire syllabus in detail. For comprehensive information and complete understanding of the course material, please refer to the full syllabus and associated readings.

Standards

- Some of the most important components within an embedded system are derived from specific methodologies, commonly referred to as standards
- market-specific standards
- general-purpose standards



Examples of standards implemented in embedded systems

- Consumer Electronics
 - JavaTV
 - DVB (Digital Video Broadcasting) – MHP (Multimedia Home Platform)
 - ISO/IEC 16500 DAVIC (Digital Audio Visual Council)
 - ATSC (Advanced Television Standards Committee) – DASE (Digital TV Applications Software Environment)
 - ATVEF (Advanced Television Enhancement Forum) – SMPTE (Society of Motion Picture and Television Engineers) DDE-1
 - DTVIA (Digital Television Industrial Alliance of China)
 - ARIB-BML (Association of Radio Industries and Business of Japan)
 - OCAP (OpenCable Application Forum)

Examples of standards implemented in embedded systems

- Consumer Electronics
 - OSGi (Open Services Gateway Initiative)
 - OpenTV
 - OpenTV
 - HAVi (Home Audio Video Initiative)
 - CEA (Consumer Electronics Association)

Examples of standards implemented in embedded systems

- Medical Devices
 - FDA (USA)
 - Medical Devices Directive (EU)
 - IEEE1073 Medical Device Communications
- Industrial Control
 - DICOM (Digital Imaging and Communications in Medicine)
 - Department of Commerce (USA) – Office of Microelectronics, Medical Equipment and Instrumentation
 - (EU) The Machinery Directive 98/37/EC
 - IEC (International Electrotechnical Commission 60204-1)
 - ISO (International Standards Organization) Standards

Examples of standards implemented in embedded systems

- Networking and Communications
 - TCP (Transmission Control Protocol)/IP (Internet Protocol)
 - PPP (Point-to-Point Protocol)
 - IEEE (Institute of Electronics and Electrical Engineers) 802.3 Ethernet
 - Cellular
- Automotive
 - GM Global
 - Ford Standards
 - FMVSS (Federal Motor Vehicle Safety Standards)
 - OPEL Engineering Material Specifications
 - Jaguar Procedures and Standards Collection
 - ISO/TS 16949 – The Harmonized Standard for the Automotive Supply Chain

Examples of standards implemented in embedded systems

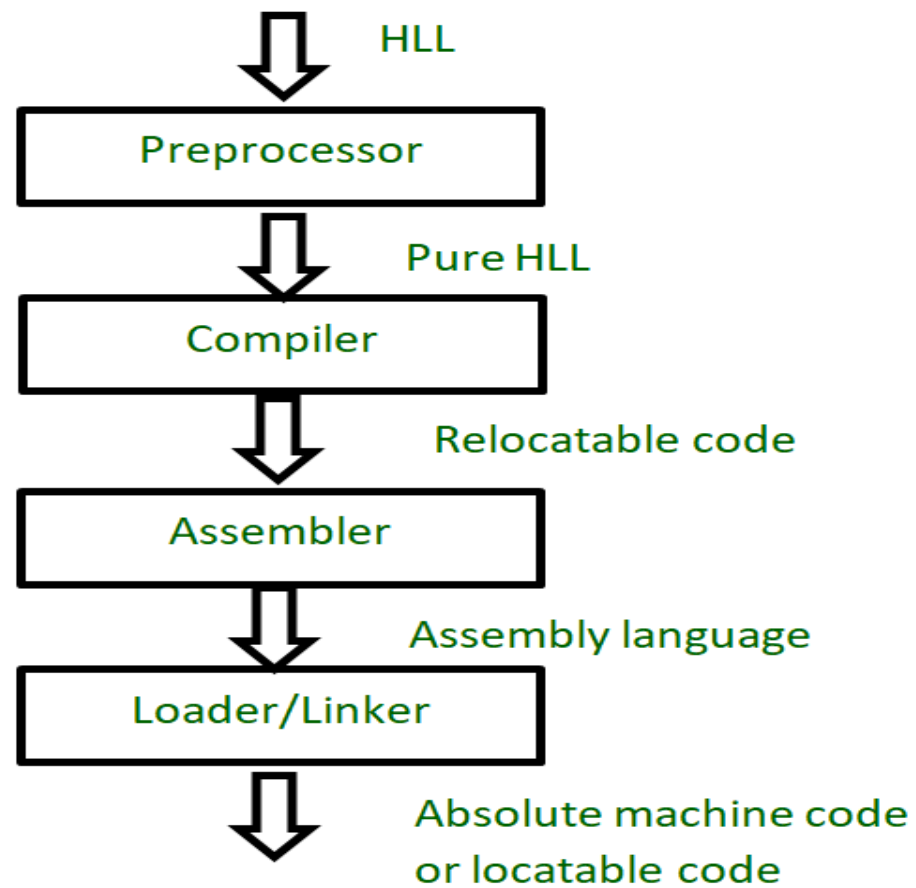
- Networking
 - HTTP (Hypertext Transfer Protocol)
 - TCP (Transmission Control Protocol)/IP (Internet Protocol)
 - IEEE (Institute of Electronics and Electrical Engineers) 802.3 Ethernet
 - Bluetooth
- Programming Languages
 - pJava (Personal Java)
 - J2ME (Java 2 Micro Edition)
 - .NET Compact Framework
 - HTML (Hyper Text Markup Language)

Examples of standards implemented in embedded systems

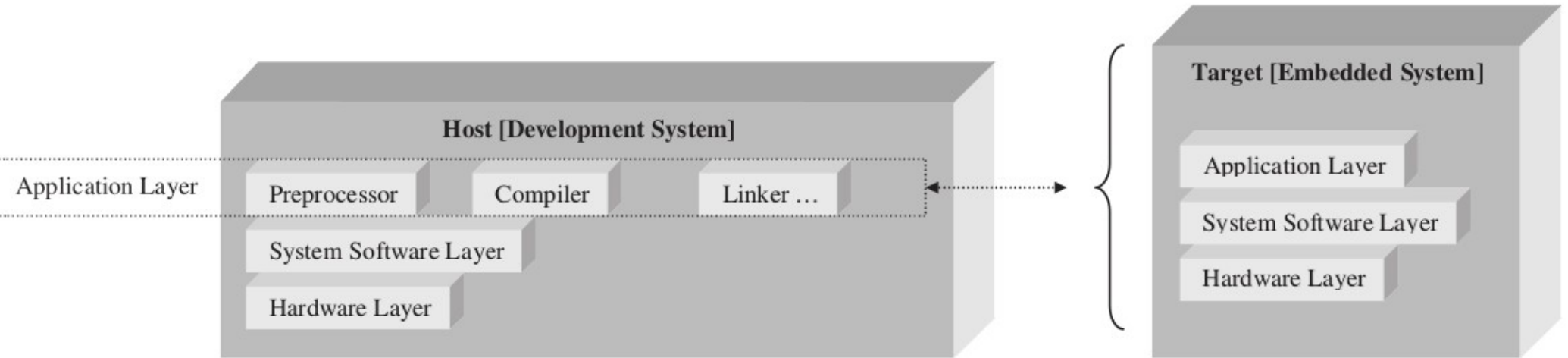
- Security
 - Netscape IETF (Internet Engineering Task Force) SSL (Secure Socket Layer) 128-bit Encryption
 - IEEE 802.10 Standards for Interoperable LAN/MAN Security (SILS)
- Quality Assurance
 - ISO 9000 Standards

Overview of Programming Languages

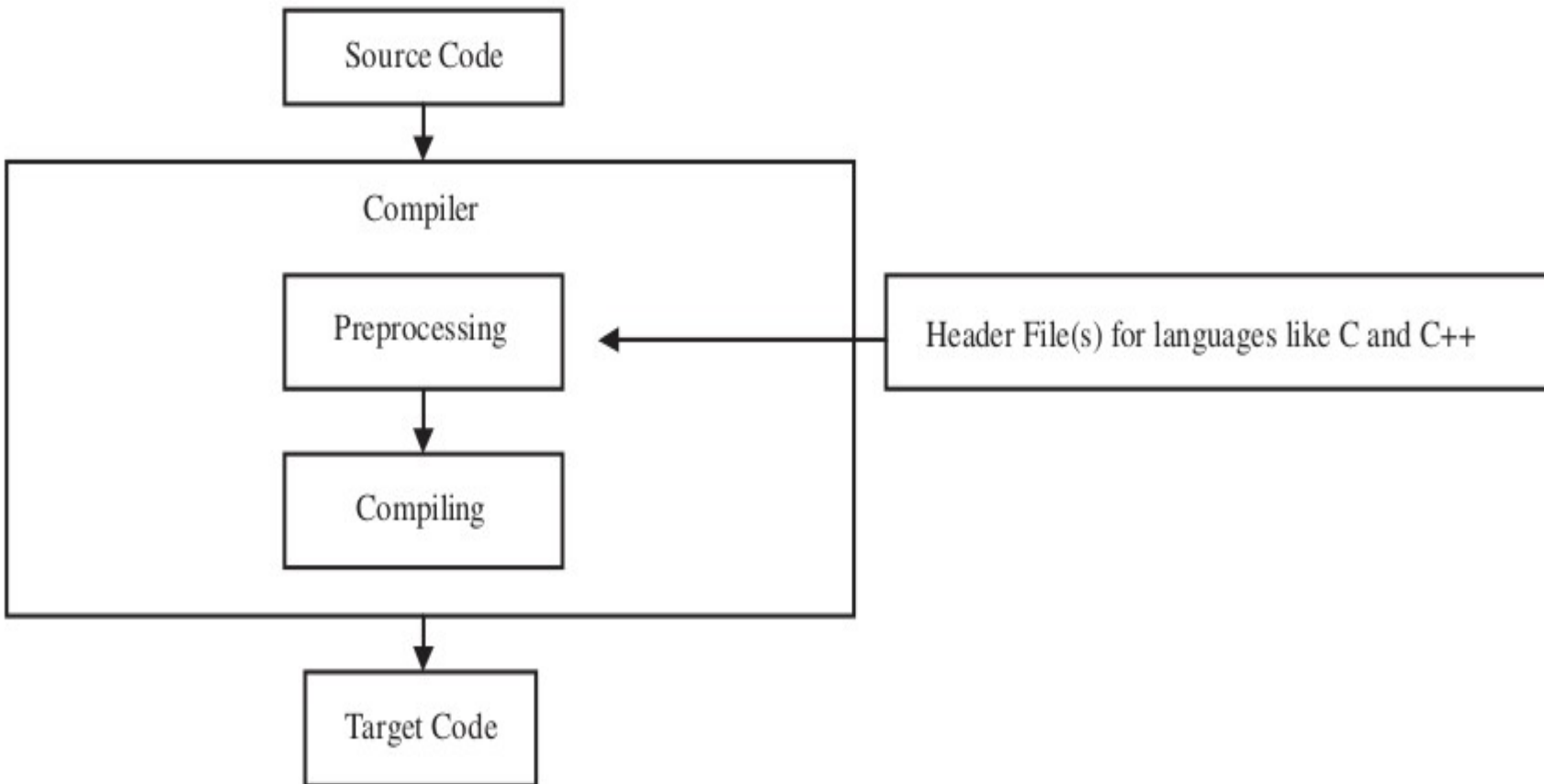
	Language	Details
1 st Generation	Machine code	Binary (0,1) and hardware dependent.
2 nd Generation	Assembly language	Hardware-dependent representing corresponding binary machine code.
3 rd Generation	HOL (high-order languages)/procedural languages	High-level languages with more English-like phrases and more transportable, such as C, Pascal, etc.
4 th Generation	VHLL (very high level languages)/non-procedural languages	“Very” high-level languages: object-oriented languages (C++, Java,...), database query languages (SQL), etc.
5 th Generation	Natural languages	Programming similar to conversational languages, typically used in artificial intelligence (AI). Still in the research and development phases in most cases—not yet applicable in mainstream embedded systems.



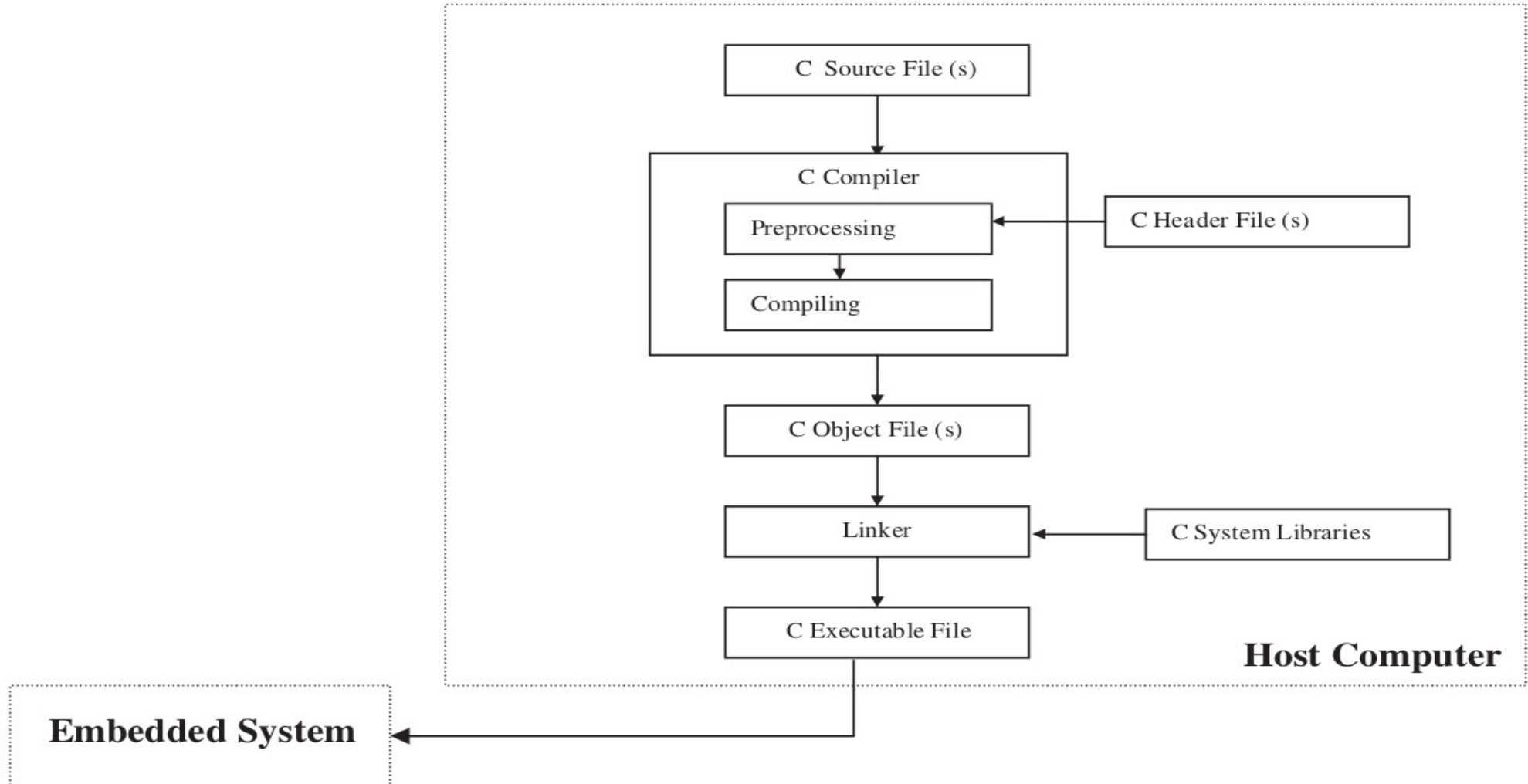
Mechanism to generate the corresponding machine code



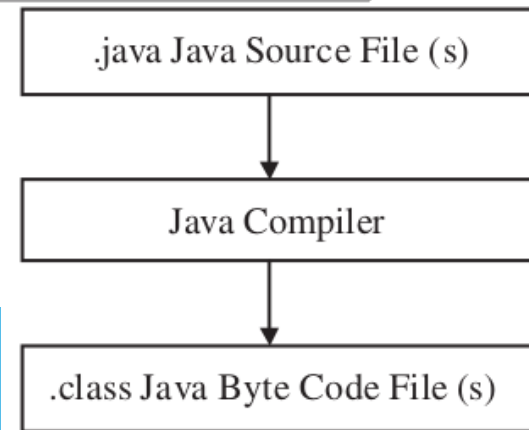
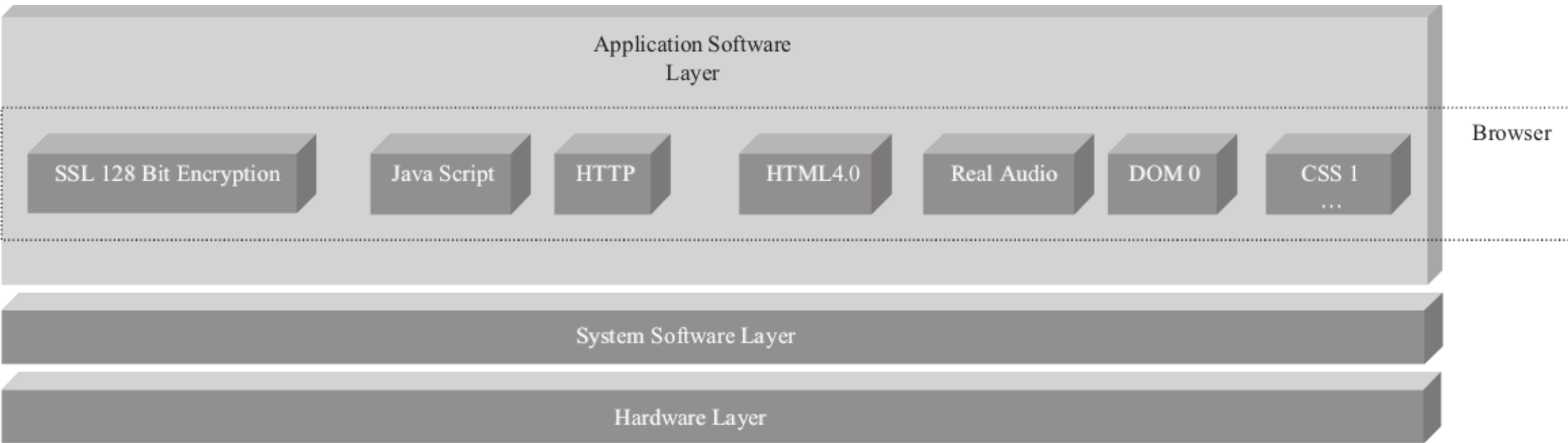
Compilation



Compilation / linking Steps

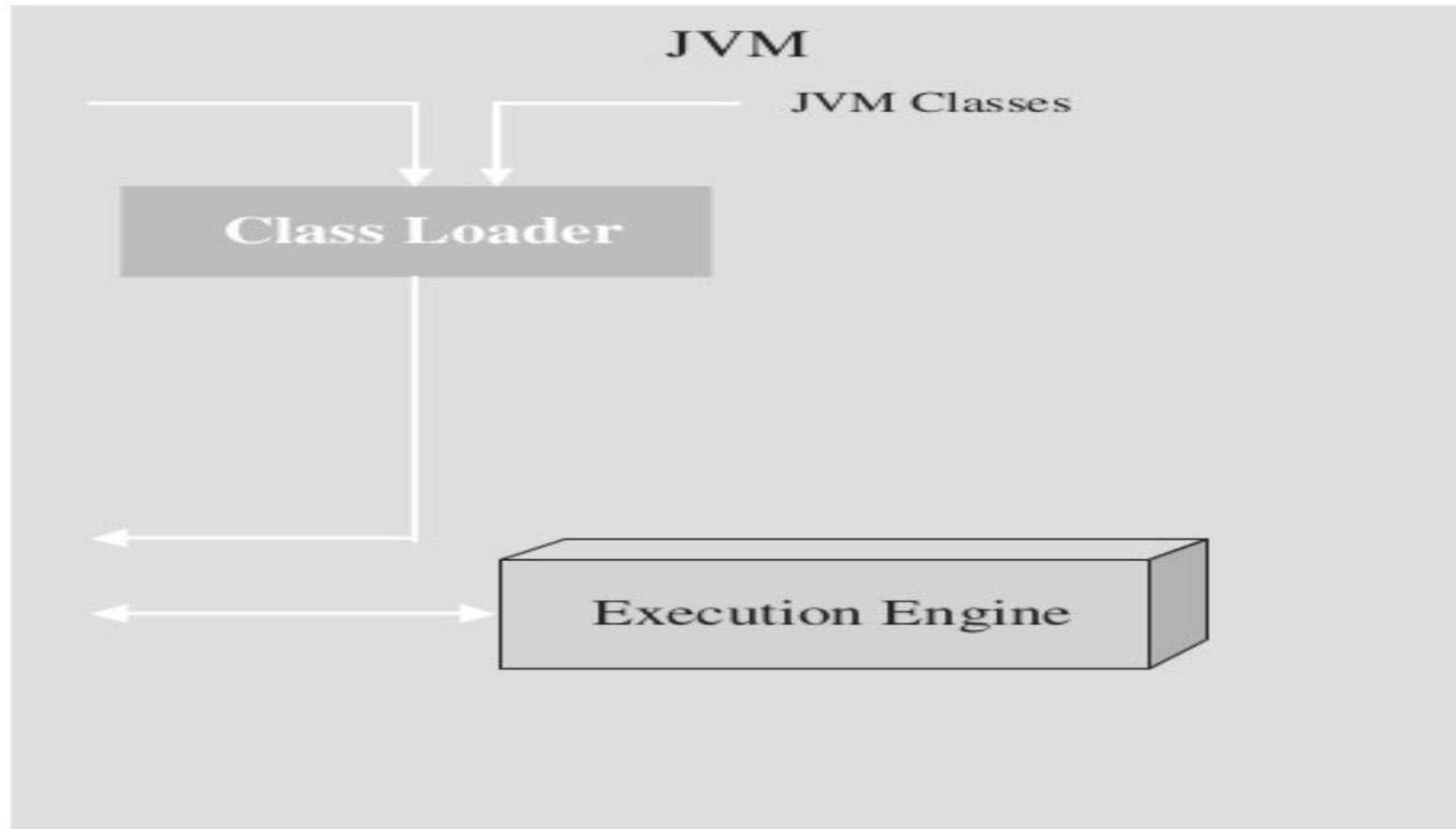


HTML and Javascript in the application layer

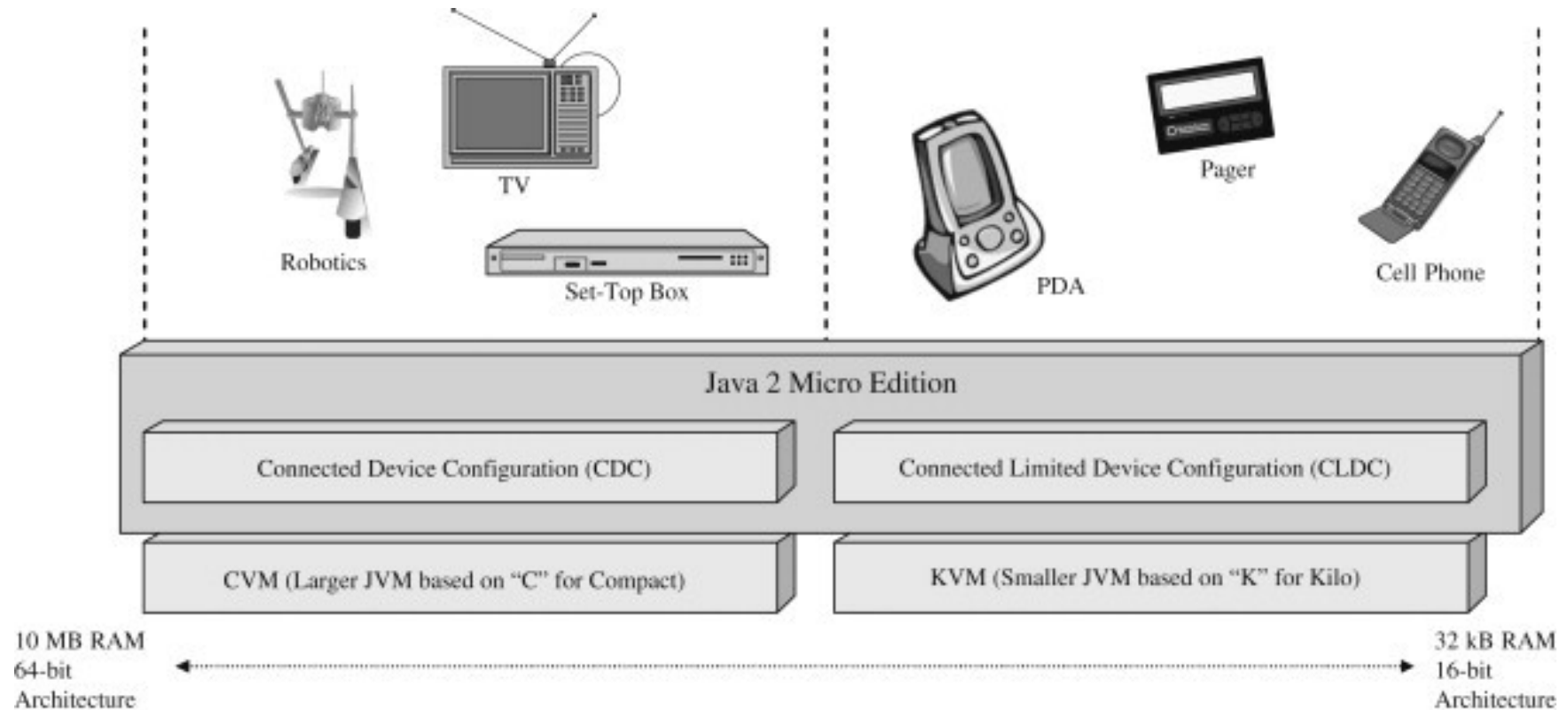


Internal JVM components

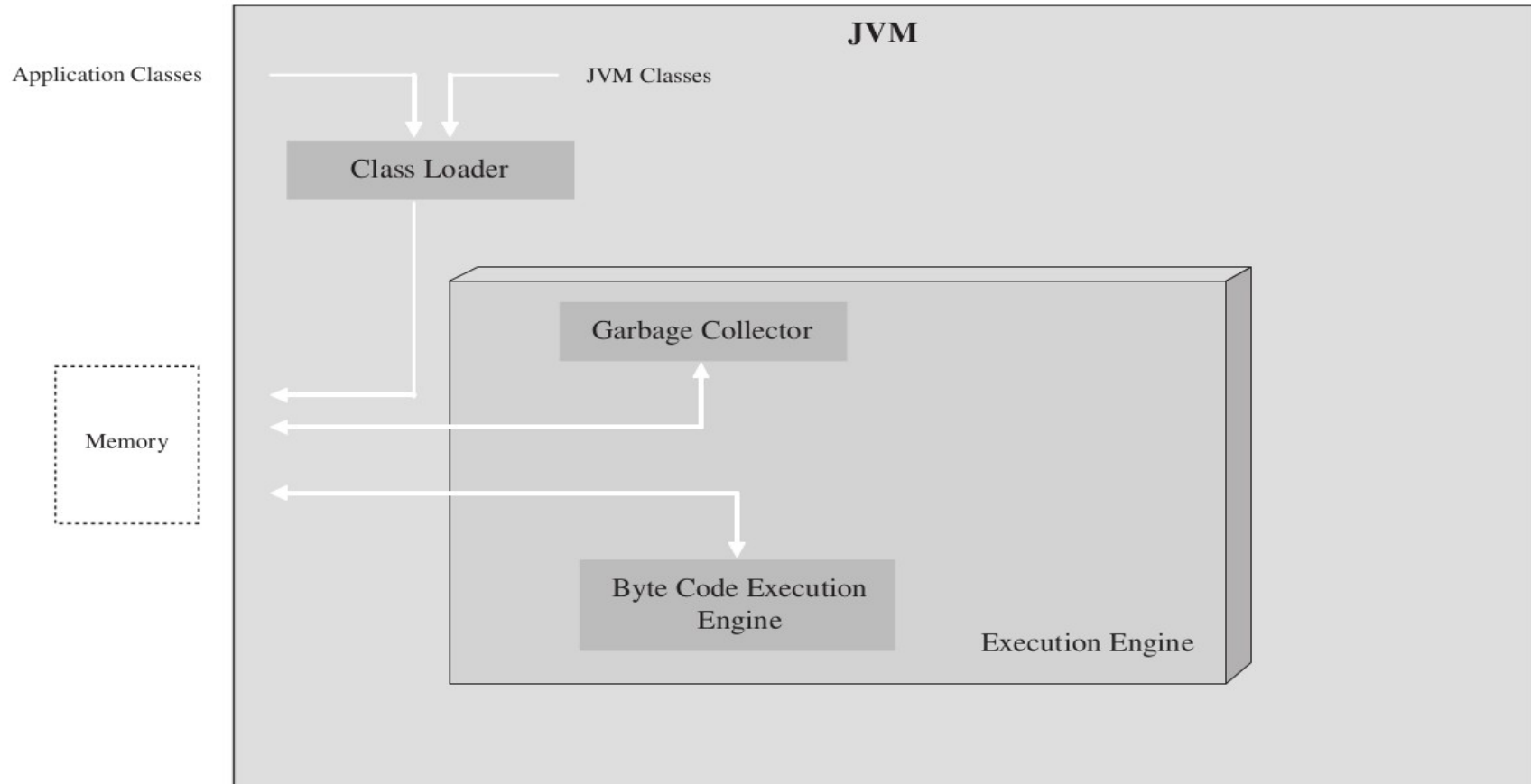
Application Classes



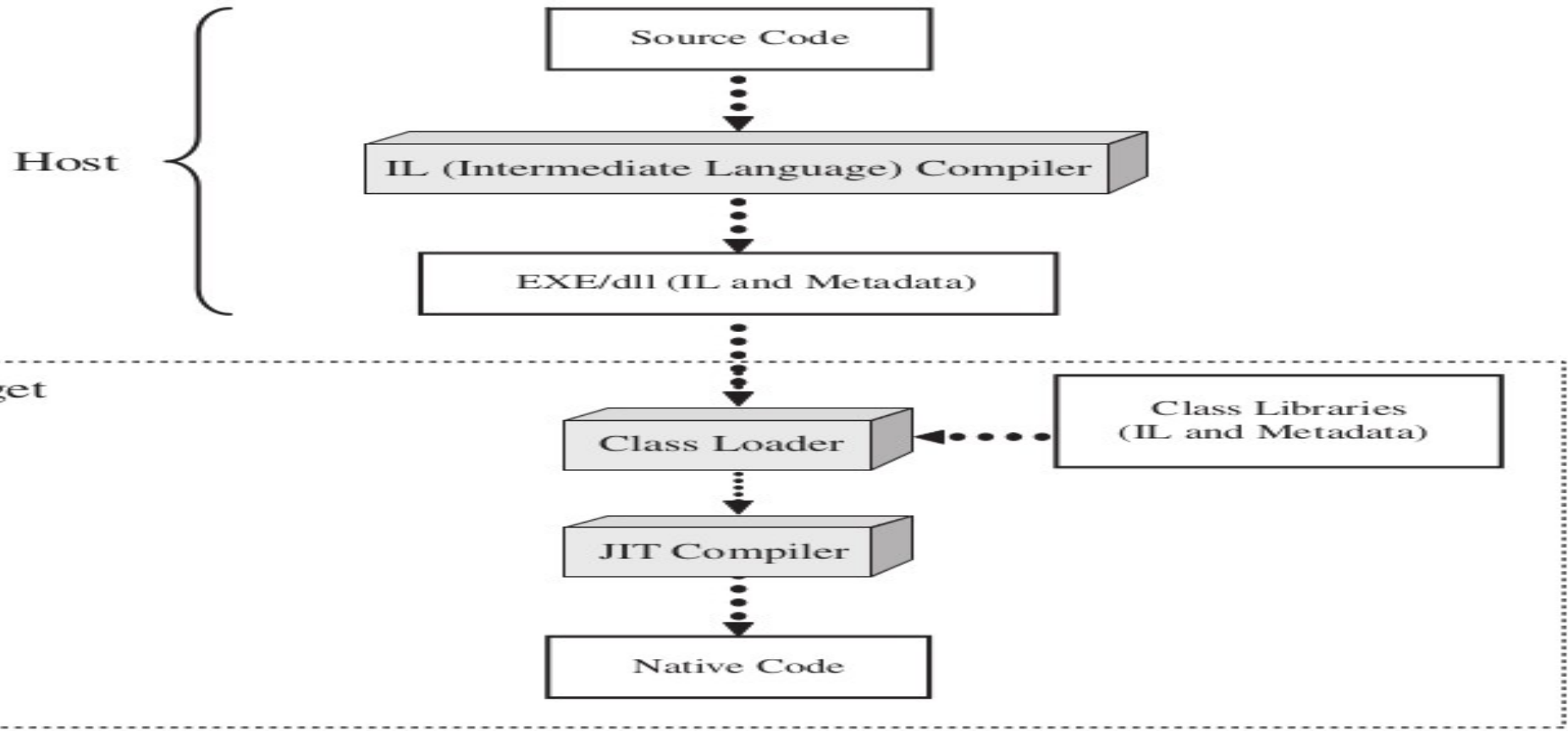
J2ME family of devices



Internal execution engine components

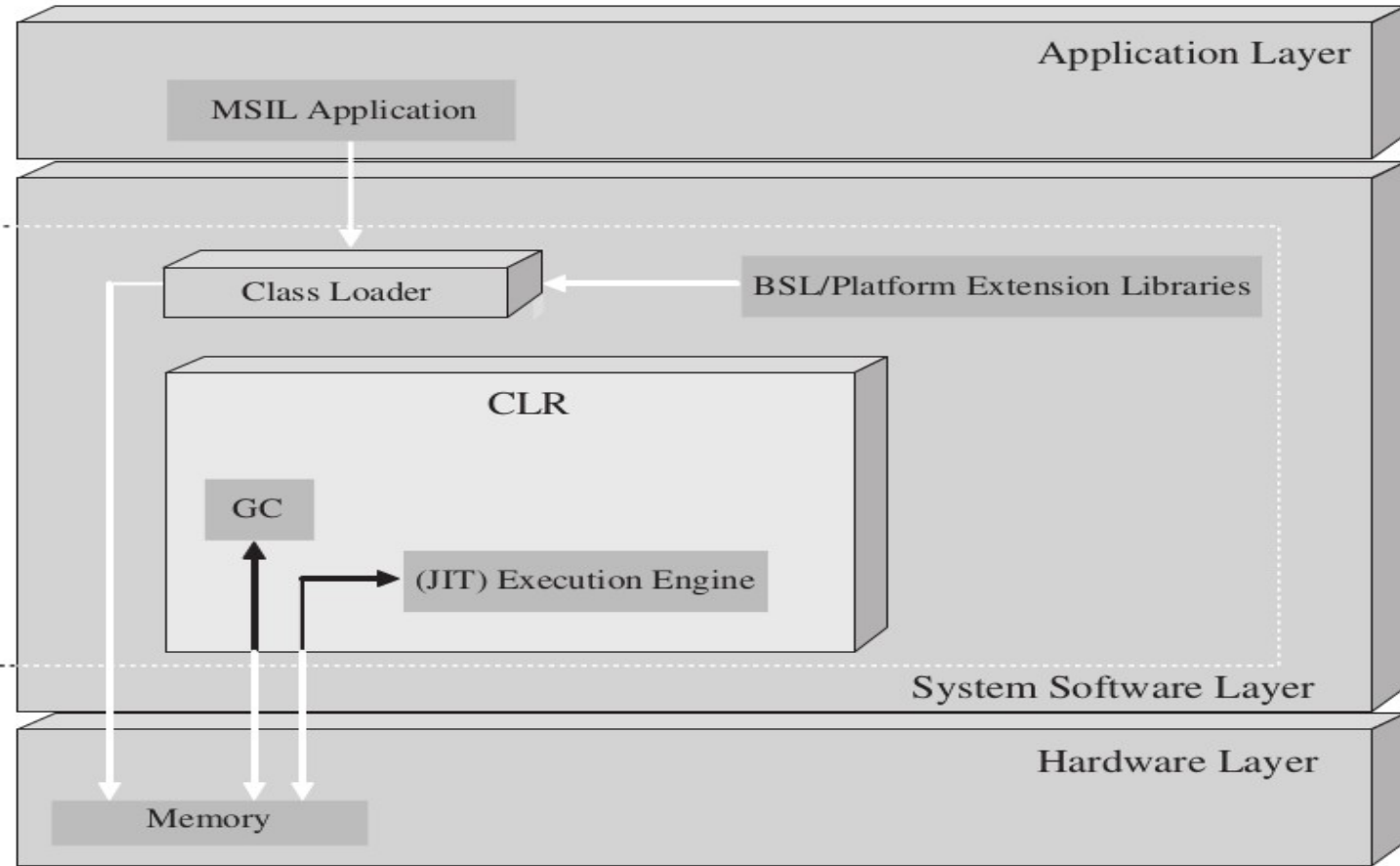


.NET Compact Framework execution model



.NET Compact Framework and the Embedded Systems Model

.NET Compact Framework



Standards and Networking

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Steps to identify required networking components

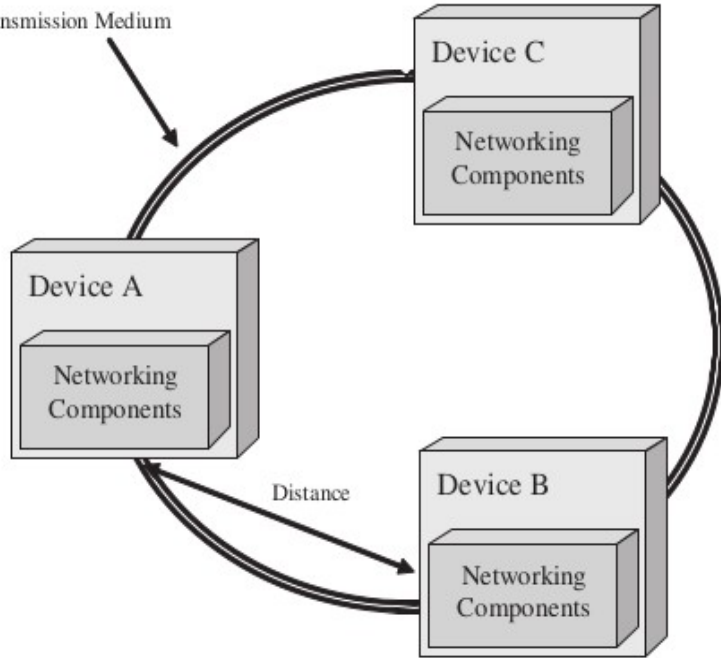
- 1) Understanding the entire network into which the device will **connect**
- 2) Using this understanding, along with a networking model, such as the OSI (Open Systems Interconnection) model discussed later in this section, to determine the device's **networking components**

Understanding the network

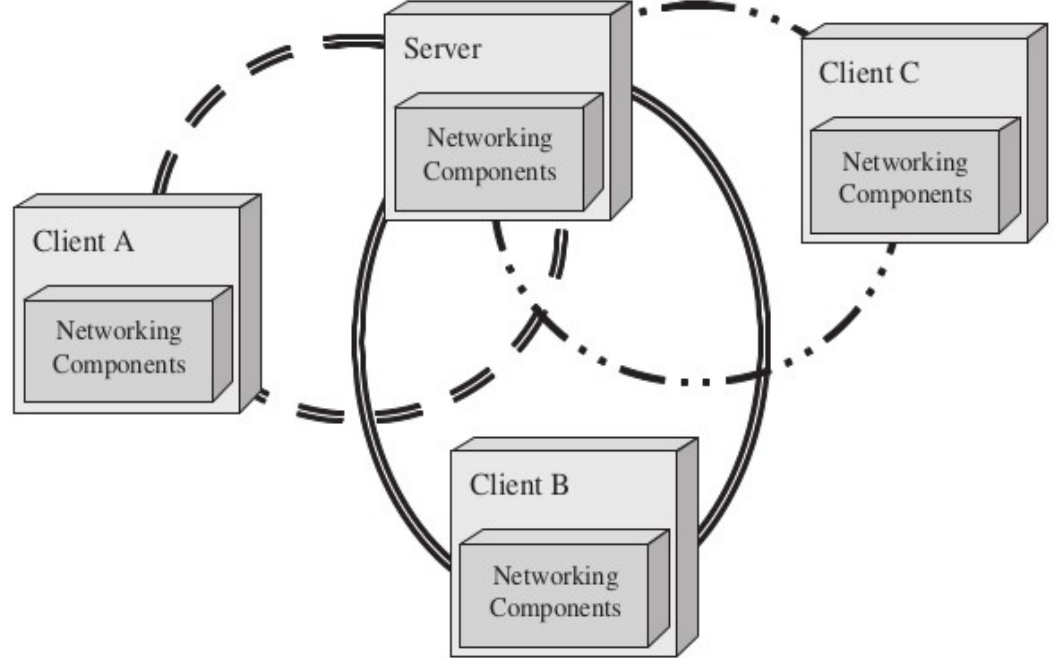
- An embedded engineer should understand three features about the entire network the device will plug into
 - the distance between connected devices
 - the physical medium connecting the embedded device to the rest of the network
 - the network's overall structure

Network block diagram

Transmission Medium



Network 1 – Peer-to-Peer Architecture



Network 2 – Client/Server Architecture

Network's Overall Structure

Distance Between Connected Devices

- LAN
- WAN
- CAN
- MAN

Physical Medium

- devices are connected with transmission mediums
 - Bound
 - cables or wires
 - “guided” mediums
 - e.g: UTP, Coaxial, Fiber Optic
 - Unbound
 - wireless connections
 - “unguided” mediums
 - e.g: Terrestrial Microwave, Satellite Microwave, Broadcast Radio, IR, Cellular Microwave

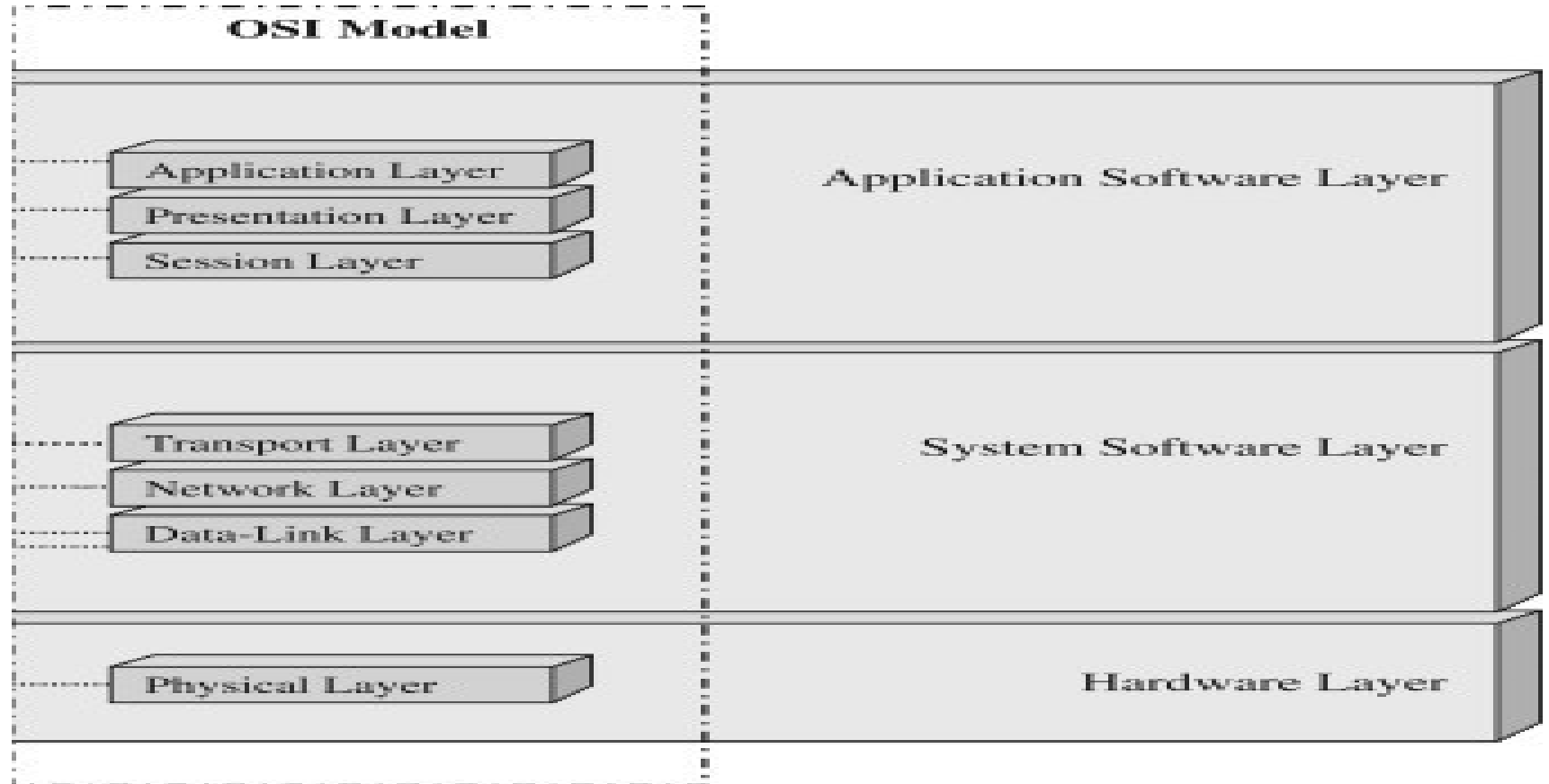
Physical Medium

- The key characteristics that differentiate all transmission mediums whether wired or wireless are
 - The type of data the medium can carry (i.e., analog or digital)
 - How much data the medium can carry (capacity)
 - How fast the medium can carry the data from source to destination (speed)
 - How far the medium can carry the data (distance)
 - How susceptible the medium is to external forces

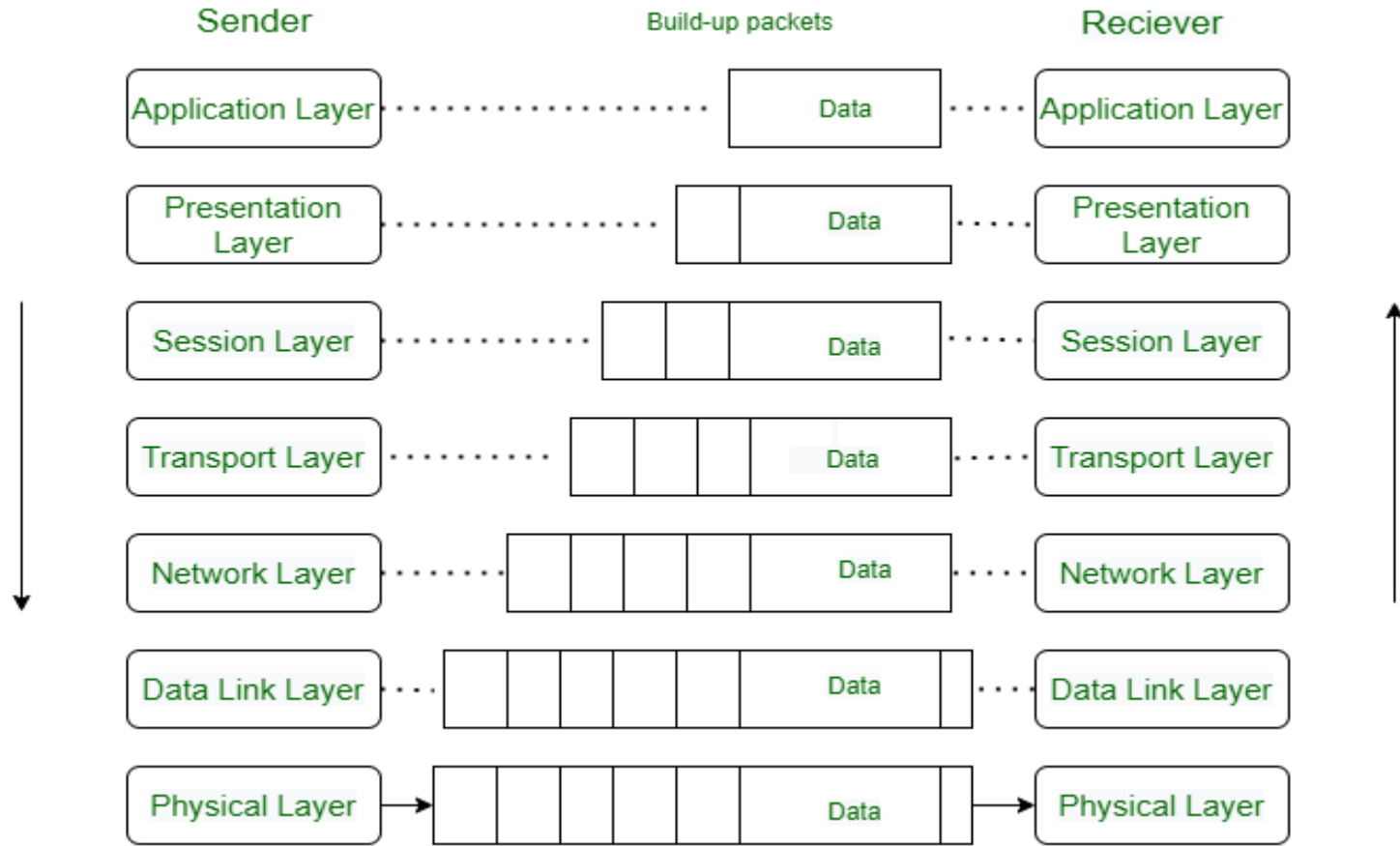
The Network Architecture

- Peer – to – Peer
- Client / Server
- Hybrid

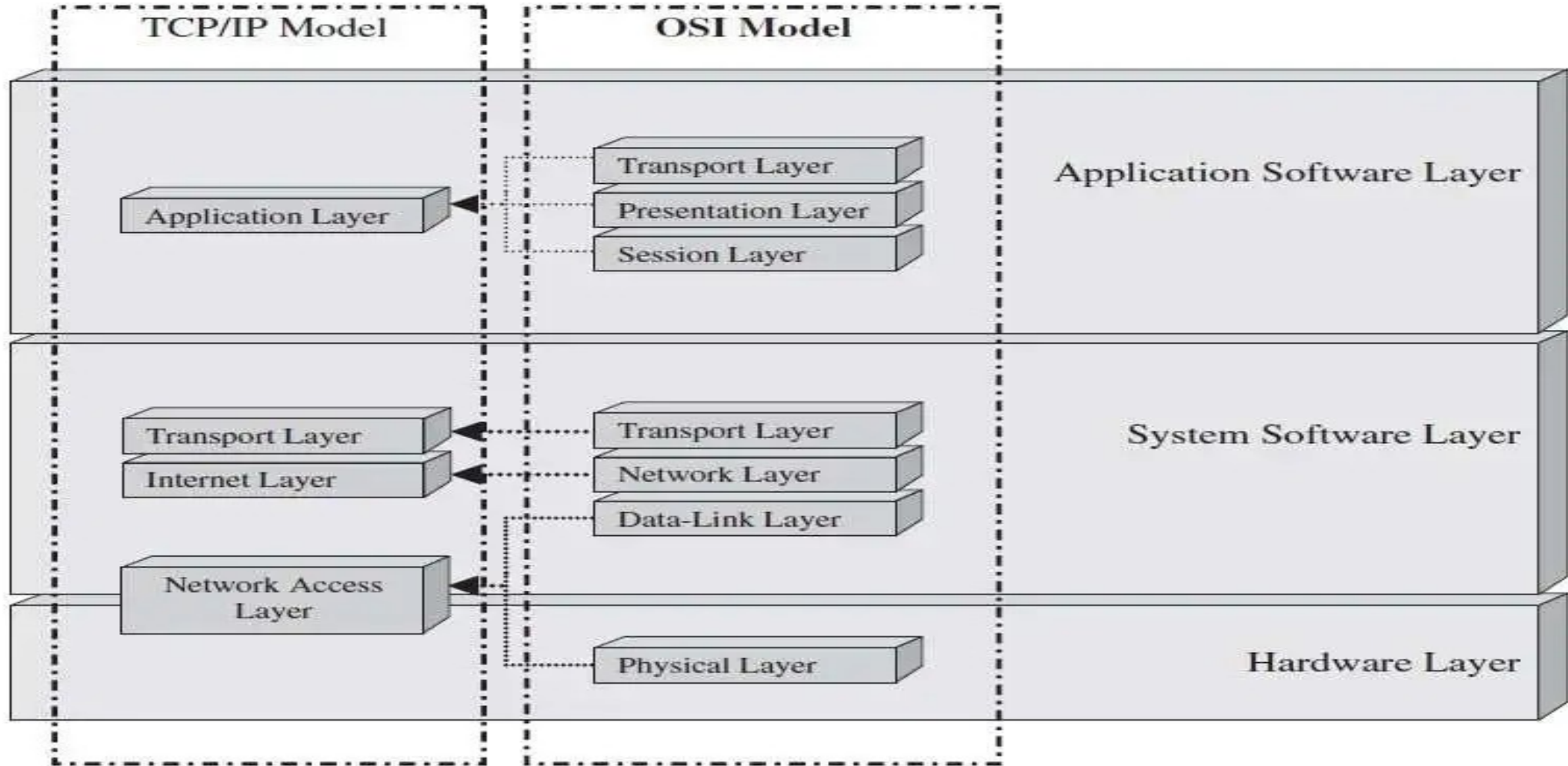
The Network Architecture



OSI model data flow diagram

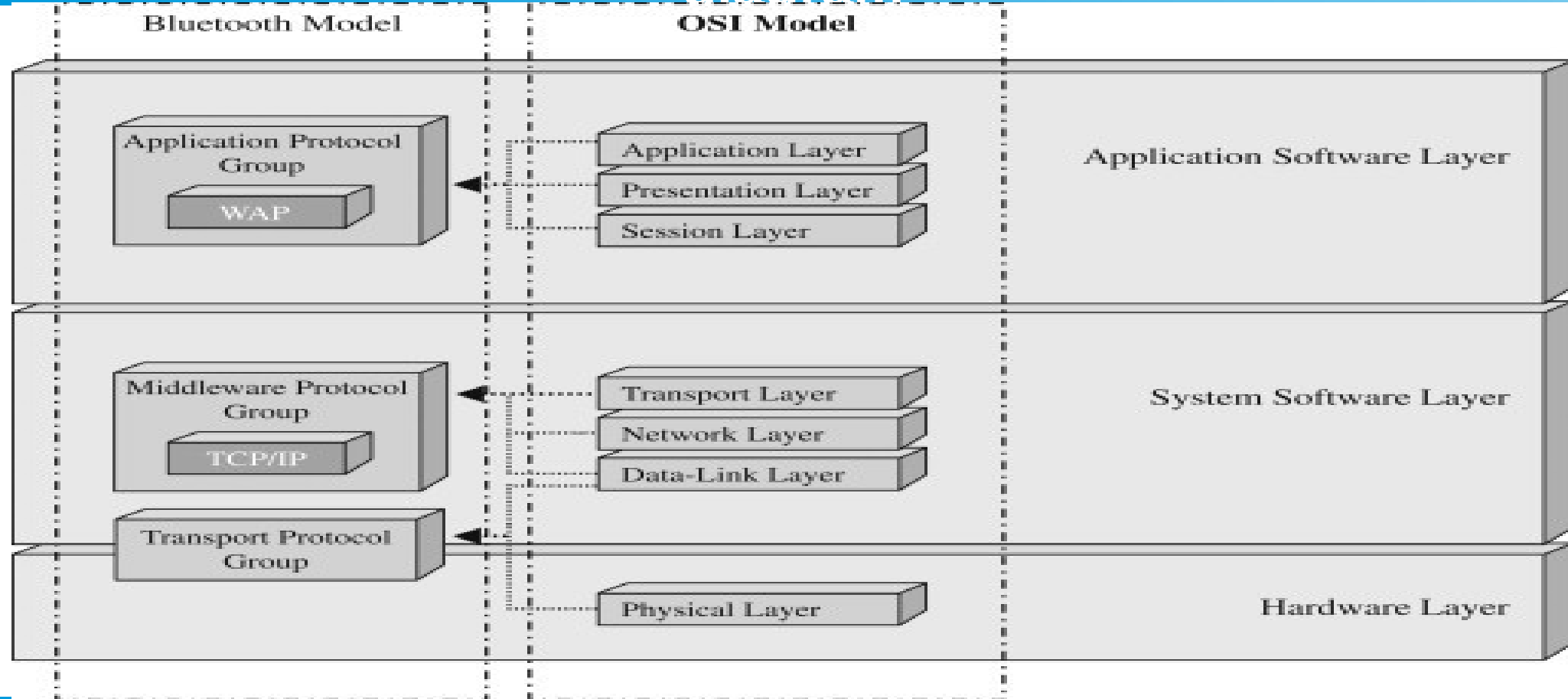


TCP/IP, OSI models and Embedded Systems Model block diagram



WAP, OSI and Embedded Systems Model block diagram

Bluetooth, OSI and Embedded Systems Model block diagram



Reference

- Tammy Noergaard (2005). Embedded Systems Architecture: A Comprehensive Guide for Engineers and Programmers. Elsevier